Measurement of a General Observable

For a normalized state, expand in an orthonormal eigenbasis of Ω:

$$|\psi\rangle = \sum_i |\omega_i\rangle\langle\omega_i \mid \psi\rangle, \qquad P(\omega) = |\langle\omega \mid \psi\rangle|^2 = \langle\psi \mid \omega\rangle\langle\omega \mid \psi\rangle = \langle\psi|P_\omega|\psi\rangle.$$

Here $P_\omega = |\omega\rangle\langle\omega|$ is the projection operator for a nondegenerate outcome. If $\langle\psi \mid \psi\rangle \neq 1$, divide by the norm:

$$P(\omega) = \frac{|\langle\omega \mid \psi\rangle|^2}{\langle\psi \mid \psi\rangle}.$$

Degenerate outcomes and state reduction

For a degenerate eigenspace, with degeneracy label λ,

$$P_\omega = \sum_\lambda |\omega,\lambda\rangle\langle\omega,\lambda|, \qquad P(\omega) = \langle\psi|P_\omega|\psi\rangle, \qquad |\psi\rangle \longmapsto \frac{P_\omega|\psi\rangle}{\|P_\omega|\psi\rangle\|}.$$

The last two expressions assume a normalized input and a nonzero outcome probability. For example, let

$$|\psi\rangle = \alpha|\omega,1\rangle + \beta|\omega,2\rangle + \gamma|\omega_3\rangle, \quad |\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1, \quad \omega_3 \neq \omega.$$

Then

$$\begin{aligned} P_\omega &= |\omega,1\rangle\langle\omega,1| + |\omega,2\rangle\langle\omega,2|, \\ P(\omega) &= \langle\psi|(\alpha|\omega,1\rangle + \beta|\omega,2\rangle) = |\alpha|^2 + |\beta|^2, \\ |\psi\rangle_{\text{after}} &= \frac{\alpha|\omega,1\rangle + \beta|\omega,2\rangle}{\sqrt{|\alpha|^2 + |\beta|^2}}. \end{aligned}$$

Expectation value

Probability theory gives, for a normalized state,

$$\begin{aligned} \langle\Omega\rangle &= \sum_i P(\omega_i)\omega_i = \sum_i |\langle\omega_i \mid \psi\rangle|^2 \omega_i \\ &= \sum_i \langle\psi \mid \omega_i\rangle\langle\omega_i \mid \psi\rangle\omega_i \\ &= \sum_i \langle\psi|\Omega|\omega_i\rangle\langle\omega_i \mid \psi\rangle \\ &= \langle\psi|\Omega\Big(\sum_i |\omega_i\rangle\langle\omega_i|\Big)|\psi\rangle = \langle\psi|\Omega|\psi\rangle. \end{aligned}$$

Uncertainty and Measurements as Filters

For a normalized state $|\psi\rangle$, write $\langle A\rangle = \langle\psi|A|\psi\rangle$. The uncertainty of an observable is

$$\Delta\Omega = \sqrt{\langle(\Omega - \langle\Omega\rangle)^2\rangle}.$$

Expanding the square gives

$$\begin{aligned}\langle(\Omega - \langle\Omega\rangle)^2\rangle &= \langle\Omega^2 - 2\langle\Omega\rangle\Omega + \langle\Omega\rangle^2 I\rangle \\ &= \langle\Omega^2\rangle - 2\langle\Omega\rangle^2 + \langle\Omega\rangle^2 \\ &= \langle\Omega^2\rangle - \langle\Omega\rangle^2,\end{aligned}$$

so $\Delta\Omega = \sqrt{\langle\Omega^2\rangle - \langle\Omega\rangle^2}$.

A single filter

$$|\psi\rangle \longrightarrow \boxed{\Omega} \longrightarrow |\omega\rangle$$

$$P(\omega) = \langle\psi|P_\omega|\psi\rangle$$

The output $|\omega\rangle$ denotes the normalized selected state; the simple ket notation assumes a nondegenerate outcome.

Two successive filters

$$|\psi\rangle \longrightarrow \boxed{\Omega} \longrightarrow |\omega\rangle \longrightarrow \boxed{\Lambda} \longrightarrow |\lambda\rangle$$

The first probability is $P(\omega) = \langle\psi|P_\omega|\psi\rangle$; the conditional probability for the second result is $P(\lambda \mid \omega) = \langle\omega|P_\lambda|\omega\rangle$. When can the second filter leave the first eigenstate unchanged, as in

$$|\psi\rangle \xrightarrow{\Omega} |\omega\rangle \xrightarrow{\Lambda} |\omega\rangle \ ?$$

For simultaneous eigenstates,

$$\Omega|\omega,\lambda\rangle = \omega|\omega,\lambda\rangle, \qquad \Lambda|\omega,\lambda\rangle = \lambda|\omega,\lambda\rangle, \qquad [\Omega,\Lambda]\,|\omega,\lambda\rangle = 0.$$

If a complete common eigenbasis exists, $[\Omega,\Lambda] = 0$: the observables are compatible. Noncommuting observables are incompatible.

Clarification. Commutation is the finite-dimensional criterion used here. For unbounded operators, domains and the relevant spectral commutation conditions must also be specified.

Infinitesimal Translations

Concept	Classical mechanics	Quantum mechanics
Translation	$x \mapsto x + \epsilon$, $p \mapsto p$	$\langle X \rangle \mapsto \langle X \rangle + \epsilon$ $\langle P \rangle \mapsto \langle P \rangle$
Translation invariance	$\mathcal{H}$ unchanged	$\langle H \rangle$ unchanged
Conservation law	$\dot{p} = 0$	$\mathrm{d}\langle P \rangle / \mathrm{d}t = 0$

For any normalized state, let $|\psi_\epsilon\rangle = T(\epsilon)|\psi\rangle$. A spatial translation must satisfy

$$\begin{aligned}\langle \psi_\epsilon | X | \psi_\epsilon \rangle &= \langle \psi | X | \psi \rangle + \epsilon, \\ \langle \psi_\epsilon | P | \psi_\epsilon \rangle &= \langle \psi | P | \psi \rangle.\end{aligned}$$

Equivalently,

$$\begin{aligned}\langle \psi | T^\dagger X T | \psi \rangle &= \langle \psi | X | \psi \rangle + \langle \psi | \epsilon I | \psi \rangle, \\ \langle \psi | T^\dagger P T | \psi \rangle &= \langle \psi | P | \psi \rangle.\end{aligned}$$

Since these hold for every state,

$$T^\dagger X T = X + \epsilon I, \qquad T^\dagger P T = P.$$

The infinitesimal generator

Since $T(0) = I$, expand a small translation as

$$T(\epsilon) = I - \frac{\mathrm{i}\epsilon}{\hbar} G + O(\epsilon^2).$$

Norm preservation requires $T^\dagger(\epsilon) T(\epsilon) = I$. Reversing the displacement undoes the transformation:

$$T(-\epsilon) = T^{-1}(\epsilon) = T^\dagger(\epsilon).$$

Comparing the first-order expansions,

$$I + \frac{\mathrm{i}\epsilon}{\hbar} G^\dagger = I + \frac{\mathrm{i}\epsilon}{\hbar} G + O(\epsilon^2), \qquad G^\dagger = G.$$

Thus the generator of a unitary translation is Hermitian.

Momentum as the Generator of Translations

Starting from $T^\dagger X T = X + \epsilon I$ and using unitarity,

$$XT = TX + \epsilon T = TX + \epsilon I + O(\epsilon^2).$$

Substitution of $T = I - \mathrm{i}\epsilon G/\hbar + O(\epsilon^2)$ gives

$$XT - TX = -\frac{\mathrm{i}\epsilon}{\hbar}[X, G] = \epsilon I + O(\epsilon^2), \qquad [X, G] = \mathrm{i}\hbar I.$$

Likewise, $T^\dagger P T = P$ implies $[P, G] = 0$. Identifying the translation generator with momentum gives

$$G = P, \qquad \boxed{[X, P] = \mathrm{i}\hbar I.}$$

Clarification. The source identifies G with P by the usual momentum and phase convention for spatial translations. The two commutators alone do not fix an additive scalar multiple of the identity.

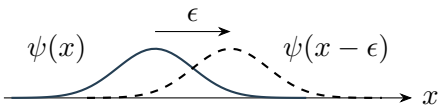
Translation of a wavefunction

Write $\psi(x) = \langle x \mid \psi \rangle$ and use $T(\epsilon)|x\rangle = |x + \epsilon\rangle$:

$$\begin{aligned} T(\epsilon)|\psi\rangle &= T(\epsilon) \int_{-\infty}^{\infty} |x\rangle\langle x \mid \psi\rangle \, \mathrm{d}x \\ &= \int_{-\infty}^{\infty} |x + \epsilon\rangle\langle x \mid \psi\rangle \, \mathrm{d}x \\ &= \int_{-\infty}^{\infty} |x'\rangle\langle x' - \epsilon \mid \psi\rangle \, \mathrm{d}x'. \end{aligned}$$

Therefore

$$\langle x|T(\epsilon)|\psi\rangle = \int \underbrace{\langle x \mid x'\rangle}_{\delta(x - x')} \langle x' - \epsilon \mid \psi\rangle \, \mathrm{d}x' = \psi(x - \epsilon).$$



Expanding both sides to first order,

$$\psi(x) - \frac{\mathrm{i}\epsilon}{\hbar}\langle x|P|\psi\rangle = \psi(x) - \epsilon\frac{\mathrm{d}\psi}{\mathrm{d}x} + O(\epsilon^2), \qquad \boxed{\langle x|P|\psi\rangle = -\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}x}\langle x \mid \psi\rangle.}$$

Infinitesimal Translation: Recap

Clarification. This source sheet is crossed through as a repeated derivation. Its substantive steps are retained below in consistent operator notation.

For a small displacement,

$$T(\epsilon) = I - \frac{\mathrm{i}\epsilon}{\hbar} G + O(\epsilon^2), \qquad T(\epsilon)|x\rangle = |x+\epsilon\rangle.$$

Reversal and norm preservation give

$$T(-\epsilon) = T^\dagger(\epsilon), \qquad T(\epsilon)T^\dagger(\epsilon) = T(-\epsilon)T(\epsilon) = I.$$

Acting on a position eigenket,

$$\begin{aligned} XT|x\rangle &= X|x+\epsilon\rangle = (x+\epsilon)|x+\epsilon\rangle, \\ TX|x\rangle &= x|x+\epsilon\rangle, \\ (XT - TX)|x\rangle &= \epsilon T|x\rangle. \end{aligned}$$

Completeness gives $T^\dagger X T = X + \epsilon I$ and

$$XT = TX + \epsilon T = TX + \epsilon I + O(\epsilon^2), \quad [X, G] = \mathrm{i}\hbar I.$$

The momentum is unchanged by translation:

$$T^\dagger P T = P \quad \Longrightarrow \quad [P, G] = 0.$$

With the standard identification $G = P$, the wavefunction transformation is

$$\psi(x) = \langle x \mid \psi\rangle, \qquad \langle x|T(\epsilon)|\psi\rangle = \psi(x-\epsilon).$$

Matching the infinitesimal expansions,

$$\psi(x) - \frac{\mathrm{i}\epsilon}{\hbar}\langle x|P|\psi\rangle = \psi(x) - \epsilon\psi'(x) + O(\epsilon^2), \qquad \langle x|P|\psi\rangle = -\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}x}\langle x \mid \psi\rangle.$$

Clarification. The crossed-through calculation writes expectations in ideal position eigenkets as if their norms were one. Such kets are delta-normalized; the operator calculation above avoids assigning a finite value to $\langle x \mid x\rangle$. For an ordinary normalized wavepacket, the intended statements are $\langle X\rangle \mapsto \langle X\rangle + \epsilon$ and $\langle P\rangle \mapsto \langle P\rangle$.

Momentum Eigenstates

Let $P|p\rangle = p|p\rangle$ and $\psi_p(x) = \langle x \mid p\rangle$. In the position basis,

$$p\langle x \mid p\rangle = \langle x|P|p\rangle = -\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}x}\langle x \mid p\rangle, \qquad \frac{\mathrm{d}\psi_p}{\mathrm{d}x} = \frac{\mathrm{i}p}{\hbar}\psi_p(x).$$

Thus $\psi_p(x) = A\mathrm{e}^{\mathrm{i}px/\hbar}$.

Delta-function normalization

Use completeness in the position basis:

$$\begin{aligned}\delta(p-p') = \langle p' \mid p\rangle &= \int_{-\infty}^{\infty}\langle p' \mid x\rangle\langle x \mid p\rangle \,\mathrm{d}x \\ &= \int_{-\infty}^{\infty} A^*\mathrm{e}^{-\mathrm{i}p'x/\hbar}A\mathrm{e}^{\mathrm{i}px/\hbar}\,\mathrm{d}x \\ &= |A|^2\int_{-\infty}^{\infty}\mathrm{e}^{\mathrm{i}(p-p')x/\hbar}\,\mathrm{d}x \\ &= |A|^2\,2\pi\hbar\,\delta(p-p').\end{aligned}$$

The last line uses the distributional Fourier identity. Hence

$$|A| = \frac{1}{\sqrt{2\pi\hbar}}, \qquad A = \frac{1}{\sqrt{2\pi\hbar}} \quad \text{by phase convention,}$$

and

$$\boxed{\langle x \mid p\rangle = \frac{1}{\sqrt{2\pi\hbar}}\mathrm{e}^{\mathrm{i}px/\hbar}.}$$

Three dimensions

For $\mathbf{P}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle$,

$$\langle \mathbf{r} \mid \mathbf{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}}\mathrm{e}^{\mathrm{i}\mathbf{p}\cdot\mathbf{r}/\hbar}, \qquad \langle \mathbf{p}' \mid \mathbf{p}\rangle = \delta^{(3)}(\mathbf{p}-\mathbf{p}').$$

The Time-Evolution Operator

The Schrödinger equation and the definition of $U(t)$ are

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = H|\psi(t)\rangle, \qquad |\psi(t)\rangle = U(t)|\psi(0)\rangle.$$

First assume a time-independent Hamiltonian with a nondegenerate discrete energy basis:

$$H|E\rangle = E|E\rangle, \qquad \sum_E |E\rangle\langle E| = I, \qquad |\psi(t)\rangle = \sum_E |E\rangle\langle E \mid \psi(t)\rangle = \sum_E a_E(t)|E\rangle.$$

Substitution yields

$$\begin{aligned} 0 &= \Big(\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t} - H\Big)|\psi(t)\rangle \\ &= \sum_E \Big(\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t} - H\Big)a_E(t)|E\rangle \\ &= \sum_E (\mathrm{i}\hbar\dot{a}_E - Ea_E)|E\rangle. \end{aligned}$$

Independence of the basis vectors gives

$$\mathrm{i}\hbar\dot{a}_E = Ea_E, \qquad a_E(t) = a_E(0)\mathrm{e}^{-\mathrm{i}Et/\hbar}, \qquad \langle E \mid \psi(t)\rangle = \langle E \mid \psi(0)\rangle\mathrm{e}^{-\mathrm{i}Et/\hbar}.$$

Consequently,

$$\begin{aligned} |\psi(t)\rangle &= \sum_E |E\rangle\langle E \mid \psi(0)\rangle\mathrm{e}^{-\mathrm{i}Et/\hbar}, \\ U(t) &= \sum_E |E\rangle\langle E|\mathrm{e}^{-\mathrm{i}Et/\hbar}. \end{aligned}$$

For degeneracy label α, replace this by

$$U(t) = \sum_{E,\alpha} |E,\alpha\rangle\langle E,\alpha|\mathrm{e}^{-\mathrm{i}Et/\hbar}.$$

A continuous part of the spectrum requires the corresponding integral over E, with its proper spectral normalization.

Stationary states

An energy eigenstate acquires only a phase, $|E(t)\rangle = \mathrm{e}^{-\mathrm{i}Et/\hbar}|E\rangle$. Thus a measurement in a fixed observable basis has time-independent probabilities:

$$|\langle\omega \mid E(t)\rangle|^2 = \left|\langle\omega \mid E\rangle\mathrm{e}^{-\mathrm{i}Et/\hbar}\right|^2 = |\langle\omega \mid E\rangle|^2 .$$

Propagators and the Choice of Representation

Since $H|E\rangle = E|E\rangle$, the spectral expression for U is

$$U(t) = \sum_E \mathrm{e}^{-\mathrm{i}Et/\hbar}|E\rangle\langle E| = \sum_E \mathrm{e}^{-\mathrm{i}Ht/\hbar}|E\rangle\langle E| = \mathrm{e}^{-\mathrm{i}Ht/\hbar}.$$

In the position basis,

$$\begin{aligned}\psi(x,t) &= \langle x \mid \psi(t)\rangle = \langle x|U(t)|\psi(0)\rangle \\ &= \int_{-\infty}^{\infty} \langle x|U(t)|x'\rangle\psi(x',0)\,\mathrm{d}x', \\ K(x,x';t) &= \langle x|U(t)|x'\rangle = \sum_E \mathrm{e}^{-\mathrm{i}Et/\hbar}\langle x \mid E\rangle\langle E \mid x'\rangle \\ &= \sum_E \mathrm{e}^{-\mathrm{i}Et/\hbar}\psi_E(x)\psi_E^*(x').\end{aligned}$$

Position in the momentum representation

To solve $H|E\rangle = E|E\rangle$ for $H = P^2/(2m) + V(X)$, choose a convenient basis. Using $\langle p \mid x\rangle = (2\pi\hbar)^{-1/2}\mathrm{e}^{-\mathrm{i}px/\hbar}$,

$$\begin{aligned}\langle p|X|\psi\rangle &= \int_{-\infty}^{\infty} \langle p \mid x\rangle\langle x|X|\psi\rangle\,\mathrm{d}x \\ &= \int_{-\infty}^{\infty} \frac{x\mathrm{e}^{-\mathrm{i}px/\hbar}}{\sqrt{2\pi\hbar}}\langle x \mid \psi\rangle\,\mathrm{d}x \\ &= \mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}p}\int_{-\infty}^{\infty} \frac{\mathrm{e}^{-\mathrm{i}px/\hbar}}{\sqrt{2\pi\hbar}}\langle x \mid \psi\rangle\,\mathrm{d}x \\ &= \mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}p}\int \langle p \mid x\rangle\langle x \mid \psi\rangle\,\mathrm{d}x = \mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}p}\langle p \mid \psi\rangle.\end{aligned}$$

The energy eigenvalue equation in either basis

Writing $\psi_E(x) = \langle x \mid E\rangle$ and $\phi_E(p) = \langle p \mid E\rangle$,

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)\right]\psi_E(x) = E\psi_E(x),$$

whereas in momentum space,

$$\left[\frac{p^2}{2m} + V\left(\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}p}\right)\right]\phi_E(p) = E\phi_E(p).$$

The last operator expression is understood through the appropriate operator calculus or Fourier kernel for the chosen potential.

Examples and Fourier Transformation

Example 1: a potential simple in position space

For the source's potential $V(X) = 1/\cosh^2 X$, write

$$H = \frac{P^2}{2m} + \frac{1}{\cosh^2 X}.$$

The position-space equation is an ordinary second-order differential equation:

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{\cosh^2 x}\right]\psi_E(x) = E\psi_E(x).$$

In momentum space it becomes

$$\left[\frac{p^2}{2m} + \frac{1}{\cosh^2\left(\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}p}\right)}\right]\phi_E(p) = E\phi_E(p).$$

Formally expanding $1/\cosh^2$ as a Taylor series produces an infinite-order differential operator, so the position representation is more convenient. The argument and coefficient scales are suppressed in the source's schematic potential.

Example 2: a constant force

Let $H = P^2/(2m) - fX$, with f constant. Then

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} - fx\right]\psi_E(x) = E\psi_E(x),$$
$$\left[\frac{p^2}{2m} - \mathrm{i}\hbar f\frac{\mathrm{d}}{\mathrm{d}p}\right]\phi_E(p) = E\phi_E(p).$$

Here momentum space gives the simpler first-order equation. If $\phi_E(p)$ is found but $\psi_E(x)$ is wanted, change representation.

The transformation between position and momentum

Insert momentum completeness:

$$\begin{aligned}\langle x \mid \psi\rangle &= \int_{-\infty}^{\infty}\langle x \mid p\rangle\langle p \mid \psi\rangle\ \mathrm{d}p \\ &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}\mathrm{e}^{\mathrm{i}px/\hbar}\langle p \mid \psi\rangle\ \mathrm{d}p.\end{aligned}$$

Thus the position wavefunction is the inverse Fourier transform of the momentum wavefunction. Conversely,

$$\begin{aligned}\langle p \mid \psi\rangle &= \int_{-\infty}^{\infty}\langle p \mid x\rangle\langle x \mid \psi\rangle\ \mathrm{d}x \\ &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}\mathrm{e}^{-\mathrm{i}px/\hbar}\langle x \mid \psi\rangle\ \mathrm{d}x.\end{aligned}$$

A Free Particle and the Parity Operator

For a free particle in one dimension,

$$H = \frac{P^2}{2m}, \qquad H|E\rangle = E|E\rangle, \qquad [H, P] = 0.$$

We can choose simultaneous eigenstates of H and P. In position space, $-\hbar^2\psi_E''/(2m) = E\psi_E$; equivalently, using momentum eigenstates,

$$\frac{P^2}{2m}|p\rangle = \frac{p^2}{2m}|p\rangle = E|p\rangle, \qquad p = \pm\sqrt{2mE}.$$

For each $E > 0$ there are two propagation directions:

$$|E, r\rangle = |p = \sqrt{2mE}\rangle, \qquad |E, l\rangle = |p = -\sqrt{2mE}\rangle.$$

The threshold $E = 0$ does not have two distinct momenta. Is momentum the only useful second label for the energy eigenstates?

Parity

Classically parity reverses $\mathbf{r}$ and $\mathbf{p}$. Quantum mechanically,

$$\Pi|\mathbf{r}\rangle = |-\mathbf{r}\rangle, \qquad \Pi|\mathbf{p}\rangle = |-\mathbf{p}\rangle.$$

In one dimension, insert position completeness:

$$\begin{aligned}
\Pi|\psi\rangle &= \Pi \int |x\rangle\langle x \mid \psi\rangle \, \mathrm{d}x = \int |-x\rangle\langle x \mid \psi\rangle \, \mathrm{d}x \\
&= \int |x'\rangle\langle -x' \mid \psi\rangle \, \mathrm{d}x', \\
\langle x|\Pi|\psi\rangle &= \int \underbrace{\langle x \mid x'\rangle}_{\delta(x-x')} \psi(-x') \, \mathrm{d}x' = \psi(-x).
\end{aligned}$$

The parity operator is unitary and satisfies $\Pi^2 = I$. Hence

$$\Pi^{-1} = \Pi^\dagger = \Pi, \qquad \text{parity eigenvalues } = \pm 1.$$

Since $\Pi X|x\rangle = x|-x\rangle$ whereas $X\Pi|x\rangle = -x|-x\rangle$,

$$\Pi X = -X\Pi, \quad \Pi^\dagger X \Pi = -X, \qquad \Pi^\dagger P \Pi = -P.$$

Inserting $\Pi\Pi^\dagger = I$ between powers gives

$$\Pi^\dagger X^n \Pi = (-1)^n X^n, \qquad \Pi^\dagger P^n \Pi = (-1)^n P^n.$$

Clarification. Commutation allows a common eigenbasis; a generic vector in a degenerate energy eigenspace need not itself be a momentum eigenvector.

Parity Eigenstates of the Free Particle

Since $\Pi^\dagger P^2 \Pi = P^2$, we have $[\Pi, P^2] = 0$ and, for the free-particle Hamiltonian, $[\Pi, H] = 0$. For a fixed $E > 0$, use $|r\rangle = ||p|\rangle$ and $|l\rangle = |-|p|\rangle$. The parity matrix in this two-direction basis is

$$\Pi \longleftrightarrow \begin{bmatrix} \langle r|\Pi|r\rangle & \langle r|\Pi|l\rangle \\ \langle l|\Pi|r\rangle & \langle l|\Pi|l\rangle \end{bmatrix} = \begin{bmatrix} \langle r \mid l\rangle & \langle r \mid r\rangle \\ \langle l \mid l\rangle & \langle l \mid r\rangle \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Its eigenvalues and normalized coefficient vectors are

$$+1: \quad \frac{e^{i\alpha}}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad -1: \quad \frac{e^{i\beta}}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Thus

$$|+\rangle = \frac{e^{i\alpha}}{\sqrt{2}}(||p|\rangle + |-|p|\rangle), \qquad |-\rangle = \frac{e^{i\beta}}{\sqrt{2}}(||p|\rangle - |-|p|\rangle).$$

These are generalized eigenstates; the two-component normalization here is the relative normalization of the two channels at fixed $|p|$.

Position-space wavefunctions

Using the plane-wave convention of note 26,

$$\begin{aligned} \langle x \mid +\rangle &= \frac{e^{i\alpha}}{\sqrt{2}}(\langle x \mid |p|\rangle + \langle x \mid -|p|\rangle) \\ &= \frac{e^{i\alpha}}{\sqrt{4\pi\hbar}}(e^{i|p|x/\hbar} + e^{-i|p|x/\hbar}) = \frac{e^{i\alpha}}{\sqrt{\pi\hbar}} \cos\frac{|p|x}{\hbar}, \\ \langle x \mid -\rangle &= \frac{e^{i\beta}}{\sqrt{2}}(\langle x \mid |p|\rangle - \langle x \mid -|p|\rangle) \\ &= \frac{ie^{i\beta}}{\sqrt{\pi\hbar}} \sin\frac{|p|x}{\hbar}. \end{aligned}$$

A convenient real-wavefunction phase convention is $\alpha = 0$, $\beta = 3\pi/2$.

Clarification. The source's intermediate odd-parity line repeats a plus sign; the minus sign above is required by its eigenvector and sine result.

The Heisenberg Equation of Motion

For a unitary transformation U, there are two equivalent ways to calculate a transformed matrix element:

	Transform the states	Transform the operator
Ket	$\|\psi\rangle \mapsto U\|\psi\rangle$	$\|\psi\rangle$ fixed
Bra	$\langle\phi\| \mapsto \langle\phi\|U^\dagger$	$\langle\phi\|$ fixed
Operator	Ω fixed	$\Omega \mapsto U^\dagger \Omega U$

Both prescriptions give

$$(\langle\phi|U^\dagger)\Omega(U|\psi\rangle) = \langle\phi|U^\dagger\Omega U|\psi\rangle.$$

Either can be used according to convenience. For example, parity gives $(\langle x'|\Pi^\dagger)X(\Pi|x\rangle) = \langle x'|\Pi^\dagger X\Pi|x\rangle$.

Two pictures of time evolution

In the Schrödinger picture, $|\psi(t)\rangle = U(t)|\psi(0)\rangle$, while an operator without explicit time dependence remains fixed. In the Heisenberg picture the state is fixed at $|\psi(0)\rangle$ and the operator evolves:

$$\Omega_H(t) = U^\dagger(t)\Omega U(t).$$

The Schrödinger equation for arbitrary initial states implies

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}U(t)|\psi(0)\rangle = HU(t)|\psi(0)\rangle, \qquad \mathrm{i}\hbar\frac{\mathrm{d}U}{\mathrm{d}t} = HU.$$

Differentiating $U^\dagger\Omega U$ gives

$$\boxed{\mathrm{i}\hbar\frac{\mathrm{d}\Omega_H}{\mathrm{d}t} = [\Omega_H, H_H]\,.}$$

This is useful when only expectations, rather than the full wavefunction, are needed. An explicitly time-dependent Schrödinger operator adds $\mathrm{i}\hbar\, U^\dagger(\partial\Omega/\partial t)U$ to the right side.

Comparison with Hamilton's equations

For $H = P^2/(2m) + V(X)$ and $[X, P] = \mathrm{i}\hbar I$,

$$\dot{x} = \frac{\partial\mathcal{H}}{\partial p} = \frac{p}{m}, \qquad \dot{X} = \frac{1}{\mathrm{i}\hbar}[X, H] = \frac{P}{m},$$
$$\dot{p} = -\frac{\partial\mathcal{H}}{\partial x} = -V'(x), \qquad \dot{P} = \frac{1}{\mathrm{i}\hbar}[P, H] = -V'(X).$$

The commutator identities used here are $[X, f(P)] = \mathrm{i}\hbar f'(P)$ and $[P, g(X)] = -\mathrm{i}\hbar g'(X)$, for functions and domains where these formal derivatives are valid.

Free-Particle Motion and Wavepacket Moments

Expectation values can be computed without first finding the evolving state:

$$\langle\psi(t)|\Omega|\psi(t)\rangle = \langle\psi(0)|\Omega_H(t)|\psi(0)\rangle.$$

For a free particle, the Heisenberg equations are

$$\frac{\mathrm{d}P(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\Big[P(t), P(t)^2/(2m)\Big] = 0,$$
$$\frac{\mathrm{d}X(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\Big[X(t), P(t)^2/(2m)\Big] = \frac{P(t)}{m}.$$

Equal-time commutators are invariant under the unitary change of picture:

$$\begin{aligned}[X(t), P(t)] &= U^\dagger X U U^\dagger P U - U^\dagger P U U^\dagger X U \\ &= U^\dagger\,[X, P]\,U = \mathrm{i}\hbar I.\end{aligned}$$

Also $[X, P^2] = [X, P]\,P + P\,[X, P] = 2\mathrm{i}\hbar P$, and more generally $[X, f(P)] = \mathrm{i}\hbar f'(P)$. Integration yields

$$P(t) = P(0), \qquad P(t)^2 = P(0)^2,$$
$$X(t) = X(0) + \frac{t}{m}P(0),$$
$$X(t)^2 = X(0)^2 + \frac{t}{m}\big[X(0)P(0) + P(0)X(0)\big] + \frac{t^2}{m^2}P(0)^2.$$

Therefore

$$\langle P(t)\rangle = \langle P(0)\rangle, \qquad \langle X(t)\rangle = \langle X(0)\rangle + \frac{\langle P(0)\rangle}{m}t.$$

For a momentum-labelled energy eigenstate, $\langle P\rangle = \pm\sqrt{2mE}$ formally; normalizable wavepackets have ordinary finite moments.

Exercise retained from the source

Start from the unnormalized Gaussian wavepacket

$$\langle x \mid \psi_u\rangle = \mathrm{e}^{\mathrm{i}p_0 x/\hbar}\mathrm{e}^{-x^2/(2\Delta^2)}, \qquad |\psi\rangle = \frac{|\psi_u\rangle}{\sqrt{\langle\psi_u \mid \psi_u\rangle}}.$$

Calculate $\langle X(0)\rangle$, $\langle X(0)^2\rangle$, $\langle P(0)\rangle$, and $\langle P(0)^2\rangle$, then use the operator solutions to find their time-dependent counterparts and

$$\Delta X(t)\Delta P(t) = \sqrt{\langle X(t)^2\rangle - \langle X(t)\rangle^2}\sqrt{\langle P(t)^2\rangle - \langle P(t)\rangle^2}.$$

Clarification. The handwritten cross term $2X(0)P(0)t/m$ in $X(t)^2$ is replaced by the symmetrized product because $X(0)$ and $P(0)$ do not commute.

The Parity Selection Rule

Let states and operators have definite parity:

$$\Pi|\psi_\alpha\rangle = (-1)^\alpha|\psi_\alpha\rangle, \qquad \Pi^\dagger O_\beta \Pi = (-1)^\beta O_\beta, \qquad \alpha,\beta \in \{0,1\}.$$

Inserting the identity twice,

$$\begin{aligned}\langle\psi_\alpha|O_\beta|\psi_\gamma\rangle &= \langle\psi_\alpha|\Pi^\dagger\Pi O_\beta \Pi^\dagger\Pi|\psi_\gamma\rangle \\ &= (-1)^\alpha(-1)^\beta(-1)^\gamma\langle\psi_\alpha|O_\beta|\psi_\gamma\rangle.\end{aligned}$$

If the product of the three parities is odd, then

$$\alpha+\beta+\gamma \text{ odd} \quad\Longrightarrow\quad \langle\psi_\alpha|O_\beta|\psi_\gamma\rangle = -\langle\psi_\alpha|O_\beta|\psi_\gamma\rangle = 0.$$

Example: a Gaussian with a travelling phase

Consider again

$$\psi_u(x) = \mathrm{e}^{\mathrm{i}p_0 x/\hbar}\mathrm{e}^{-x^2/(2\Delta^2)}.$$

When $p_0 = 0$ the state is even. Both X and P are odd, so their expectations vanish by the selection rule. When $p_0 \neq 0$, the state is not a parity eigenstate, so that rule cannot be applied directly. Nevertheless, the density is even:

$$\begin{aligned}\langle X\rangle &= \frac{1}{\langle\psi_u \mid \psi_u\rangle}\int_{-\infty}^{\infty}\psi_u(x)^* x\psi_u(x)\,\mathrm{d}x \\ &= \frac{1}{\langle\psi_u \mid \psi_u\rangle}\int_{-\infty}^{\infty} x\mathrm{e}^{-x^2/\Delta^2}\,\mathrm{d}x = 0.\end{aligned}$$

Equivalently, inserting $\Pi^\dagger\Pi$ on either side of X changes x into $-x$ and $\psi_u(x)$ into $\psi_u(-x)$; their squared moduli are equal. Thus the transformed integral is $-\langle X\rangle$. For momentum, direct differentiation instead gives

$$\begin{aligned}-\mathrm{i}\hbar\psi_u'(x) &= \left(p_0 + \frac{\mathrm{i}\hbar x}{\Delta^2}\right)\psi_u(x), \\ \langle P\rangle &= p_0 + \frac{\mathrm{i}\hbar}{\Delta^2}\langle X\rangle = p_0.\end{aligned}$$

Clarification. The source ends with "similarly, $\langle P\rangle = 0$." That conclusion is valid for $p_0 = 0$ only; the travelling-phase example has $\langle P\rangle = p_0$.

Classically Allowed and Forbidden Regions

Classically the kinetic energy is $E - V(x)$. Motion is allowed where $E \geq V(x)$ and forbidden where $E < V(x)$. The source illustrates smooth and square wells, infinite walls, and a well approaching a finite value:



Locally classically allowed: oscillatory solutions

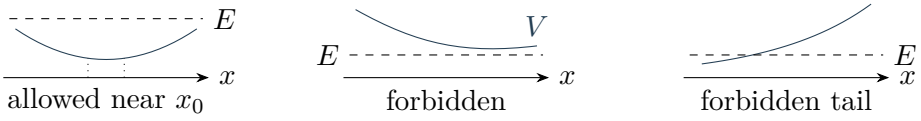
In a small interval around x_0 where the potential is approximately constant,

$$\frac{2m}{\hbar^2}[E - V(x)] \simeq k^2 \geq 0, \qquad \psi'' \simeq -k^2\psi, \qquad \psi \simeq A\mathrm{e}^{\mathrm{i}kx} + B\mathrm{e}^{-\mathrm{i}kx}.$$

Locally classically forbidden: exponential solutions

If $E < V(x)$ and V is approximately constant in the interval,

$$\frac{2m}{\hbar^2}[E - V(x)] \simeq -\kappa^2 < 0, \qquad \psi'' \simeq \kappa^2\psi, \qquad \psi \simeq A\mathrm{e}^{\kappa x} + B\mathrm{e}^{-\kappa x}, \quad \kappa > 0.$$



If $E < V(x)$ for all $x > x_0$, a physical bound-state wavefunction must exclude the growing right-tail solution and decay as $x \to +\infty$. Likewise, a forbidden left tail must decay as $x \to -\infty$. For a confining potential with $V(x) \to +\infty$ as $|x| \to \infty$, for each E there may be turning points $x_1 > x_2$ with $E < V(x)$ for $x > x_1$ and $x < x_2$; physical eigenstates decaying at both ends are bound states.

Clarification. The source writes $A(x)\mathrm{e}^{\kappa(x)x} + B(x)\mathrm{e}^{-\kappa(x)x}$. This is local-constant shorthand, not a general solution for variable κ. Where a WKB approximation applies, the tails instead have the form $\kappa(x)^{-1/2}\exp[\pm\int^x \kappa(s)\,\mathrm{d}s]$; turning points need separate treatment.

Nondegeneracy of One-Dimensional Bound States

Theorem

There is no degeneracy among bound states of the ordinary one-dimensional Schrödinger problem on a connected line, with a real scalar potential and the usual bound-state boundary conditions.

Proof exercise

The source assigns the proof as homework and supplies no proof on this page.

Clarification. The qualifications specify the standard setting of the handwritten theorem. The statement is not intended to cover disconnected domains, internal degrees of freedom, or arbitrary boundary conditions.

The Harmonic Oscillator: Factorization

The energy eigenvalue problem is

$$\left(\frac{P^2}{2m}+\frac{1}{2}m\omega^2X^2\right)|E\rangle = E|E\rangle.$$

Classically,

$$\mathcal{H}=\frac{p^2+m^2\omega^2x^2}{2m}=\frac{1}{2m}(m\omega x-\mathrm{i}p)(m\omega x+\mathrm{i}p).$$

Quantum mechanically the commutator changes this factorization:

$$\begin{aligned}\frac{1}{2m}(m\omega X-\mathrm{i}P)(m\omega X+\mathrm{i}P) &= \frac{1}{2m}\left[m^2\omega^2X^2+P^2+\mathrm{i}m\omega(XP-PX)\right]\\ &= H-\frac{\hbar\omega}{2}, \qquad [X,P]=\mathrm{i}\hbar I.\end{aligned}$$

Introduce

$$\alpha=\sqrt{\frac{m\omega}{2\hbar}}, \qquad \beta=\sqrt{\frac{1}{2m\omega\hbar}}, \qquad \alpha\beta=\frac{1}{2\hbar}.$$

The factored expression can also be written

$$\begin{aligned}H-\frac{\hbar\omega}{2} &= \hbar\omega\left[\frac{m\omega}{2\hbar}X^2+\frac{1}{2m\omega\hbar}P^2+\frac{\mathrm{i}}{2\hbar}(XP-PX)\right]\\ &= \hbar\omega\left(\sqrt{\frac{m\omega}{2\hbar}}X-\mathrm{i}\sqrt{\frac{1}{2m\omega\hbar}}P\right)\left(\sqrt{\frac{m\omega}{2\hbar}}X+\mathrm{i}\sqrt{\frac{1}{2m\omega\hbar}}P\right)\\ &= \hbar\omega(\alpha X-\mathrm{i}\beta P)(\alpha X+\mathrm{i}\beta P).\end{aligned}$$

Thus

$$\boxed{H=\hbar\omega\left[(\alpha X-\mathrm{i}\beta P)(\alpha X+\mathrm{i}\beta P)+\frac{1}{2}\right].}$$

The commutator of the two factors is

$$\begin{aligned}[\alpha X+\mathrm{i}\beta P,\alpha X-\mathrm{i}\beta P] &= [\alpha X+\mathrm{i}\beta P,\alpha X]+[\alpha X+\mathrm{i}\beta P,-\mathrm{i}\beta P]\\ &= \mathrm{i}\alpha\beta\,[P,X]-\mathrm{i}\alpha\beta\,[X,P]\\ &= 2\hbar\alpha\beta\,I=I.\end{aligned}$$

Ladder Operators and the Ground-State Energy

Define

$$a = \alpha X + \mathrm{i}\beta P, \qquad a^\dagger = \alpha X - \mathrm{i}\beta P, \qquad \left[a, a^\dagger\right] = I, \qquad H = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right).$$

For convenience, use $\widehat{H} = H/(\hbar\omega) = a^\dagger a + 1/2$ and $\widehat{H}|\epsilon\rangle = \epsilon|\epsilon\rangle$. Then

$$\left[a, \widehat{H}\right] = \left[a, a^\dagger a\right] = \left[a, a^\dagger\right] a = a,$$
$$\left[a^\dagger, \widehat{H}\right] = a^\dagger\left[a^\dagger, a\right] = -a^\dagger.$$

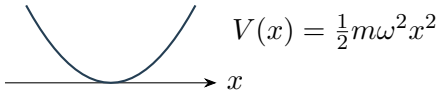
Consequently,

$$\widehat{H}a|\epsilon\rangle = (a\widehat{H} - \left[a, \widehat{H}\right])|\epsilon\rangle = (a\widehat{H} - a)|\epsilon\rangle = (\epsilon - 1)a|\epsilon\rangle,$$
$$\widehat{H}a^\dagger|\epsilon\rangle = (a^\dagger\widehat{H} - \left[a^\dagger, \widehat{H}\right])|\epsilon\rangle = (\epsilon + 1)a^\dagger|\epsilon\rangle.$$

Since this is a one-dimensional bound-state problem, nonzero states on each rung are unique up to normalization:

$$a|\epsilon\rangle = B_\epsilon|\epsilon - 1\rangle, \qquad a^\dagger|\epsilon\rangle = C_\epsilon|\epsilon + 1\rangle.$$

The oscillator is confining:



Could lowering continue indefinitely to $\epsilon - 1, \epsilon - 2, \ldots$? For every normalizable state in the Hamiltonian's domain,

$$\begin{aligned}\langle\psi|H|\psi\rangle &= \frac{1}{2m}\langle\psi|P^\dagger P|\psi\rangle + \frac{1}{2}m\omega^2\langle\psi|X^\dagger X|\psi\rangle \\ &= \frac{1}{2m}\left\|P|\psi\rangle\right\|^2 + \frac{1}{2}m\omega^2\left\|X|\psi\rangle\right\|^2 \geq 0.\end{aligned}$$

An indefinitely descending ladder would eventually have negative energy. Hence there is a lowest state annihilated by a:

$$a|\epsilon_0\rangle = 0 \implies a^\dagger a|\epsilon_0\rangle = 0 \implies \widehat{H}|\epsilon_0\rangle = \frac{1}{2}|\epsilon_0\rangle, \qquad \epsilon_0 = \frac{1}{2}.$$

The Oscillator Spectrum and Normalized Ladder States

Starting at $\epsilon_0 = 1/2$, successive raising gives

$$\frac{1}{2},\ 1+\frac{1}{2},\ 2+\frac{1}{2},\ldots, \qquad \epsilon_n = n+\frac{1}{2}, \qquad E_n = \left(n+\frac{1}{2}\right)\hbar\omega, \quad n=0,1,2,\ldots.$$

These are all the eigenvalues: any other ladder would have a lowest state annihilated by a, again with energy $\hbar\omega/2$, but this one-dimensional bound-state ground energy is nondegenerate.

Determining the ladder coefficients

Relabel $|\epsilon\rangle$ by $|n\rangle$ and write

$$a|n\rangle = B_n|n-1\rangle, \qquad \langle n|a^\dagger = B_n^*\langle n-1|.$$

Normalization gives

$$|B_n|^2\,\langle n-1\mid n-1\rangle = \langle n|a^\dagger a|n\rangle = \langle n|\widehat{H}-1/2|n\rangle = n\langle n\mid n\rangle = n,$$

so $B_n = \sqrt{n}\,\mathrm{e}^{\mathrm{i}\phi_n}$. Similarly,

$$a^\dagger|n\rangle = \sqrt{n+1}\,\mathrm{e}^{\mathrm{i}\lambda_n}|n+1\rangle.$$

Acting successively with a and $a^\dagger$,

$$a^\dagger a|n\rangle = n\mathrm{e}^{\mathrm{i}(\phi_n+\lambda_{n-1})}|n\rangle = n|n\rangle, \qquad \lambda_{n-1} = -\phi_n \pmod{2\pi}.$$

Thus $a^\dagger|n\rangle = \sqrt{n+1}\,\mathrm{e}^{-\mathrm{i}\phi_{n+1}}|n+1\rangle$.

Removing the arbitrary phases

For the first two steps,

$$a^\dagger|0\rangle = \mathrm{e}^{-\mathrm{i}\phi_1}|1\rangle = |1\rangle',$$
$$a^\dagger|1\rangle' = \sqrt{2}\,\mathrm{e}^{-\mathrm{i}(\phi_1+\phi_2)}|2\rangle = \sqrt{2}|2\rangle'.$$

In general choose $|n\rangle' = \exp[-\mathrm{i}\sum_{k=1}^{n}\phi_k]|n\rangle$. Then $a^\dagger|n-1\rangle' = \sqrt{n}|n\rangle'$. Dropping the primes for this phase convention,

$$\boxed{a|n\rangle = \sqrt{n}|n-1\rangle, \qquad a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle.}$$

Repeated raising constructs the normalized states:

$$|n\rangle = \frac{a^\dagger}{\sqrt{n}}|n-1\rangle = \frac{a^\dagger}{\sqrt{n}}\frac{a^\dagger}{\sqrt{n-1}}|n-2\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle.$$

Number Operator and Oscillator Matrix Elements

Since $a^\dagger a|n\rangle = n|n\rangle$, define the number operator

$$N = a^\dagger a, \qquad \widehat{H} = N + \frac{1}{2}.$$

The ladder-operator matrix elements are

$$\langle n'|a|n\rangle = \sqrt{n}\langle n' \mid n-1\rangle = \sqrt{n}\,\delta_{n',n-1},$$
$$\langle n'|a^\dagger|n\rangle = \sqrt{n+1}\langle n' \mid n+1\rangle = \sqrt{n+1}\,\delta_{n',n+1}.$$

Invert the definitions of a and $a^\dagger$:

$$X = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger), \qquad P = \mathrm{i}\sqrt{\frac{m\omega\hbar}{2}}(a^\dagger - a),$$

which determine $\langle n'|X|n\rangle$ and $\langle n'|P|n\rangle$. Also,

$$\langle n'|H|n\rangle = \left(n + \frac{1}{2}\right)\hbar\omega\,\delta_{n'n}.$$

Example: a matrix element of the cube of position

Powers such as X^ℓ or P^ℓ can be evaluated by successive ladder actions; moments also determine uncertainties such as $\Delta X = \sqrt{\langle X^2\rangle - \langle X\rangle^2}$. For the example $\langle 3|X^3|2\rangle$, set $c = \sqrt{\hbar/(2m\omega)}$. Then

$$X^3 = c^3(a + a^\dagger)^3 = c^3(a^3 + a^2a^\dagger + aa^\dagger a + a(a^\dagger)^2 + a^\dagger a^2 + a^\dagger aa^\dagger + (a^\dagger)^2 a + (a^\dagger)^3).$$

The order of factors matters. To connect $|2\rangle$ to $\langle 3|$, a term must contain one more raising operator than lowering operators. For example $a^\dagger aa|2\rangle$ ends on $|1\rangle$ and does not contribute. The three surviving terms give

$$(a^\dagger)^2 a|2\rangle = \sqrt{2}(a^\dagger)^2|1\rangle = 2a^\dagger|2\rangle = 2\sqrt{3}|3\rangle,$$
$$a^\dagger aa^\dagger|2\rangle = \sqrt{3}a^\dagger a|3\rangle = 3\sqrt{3}|3\rangle,$$
$$a(a^\dagger)^2|2\rangle = \sqrt{3}\sqrt{4}\,a|4\rangle = 4\sqrt{3}|3\rangle.$$

Therefore

$$\boxed{\langle 3|X^3|2\rangle = 9\sqrt{3}\left(\frac{\hbar}{2m\omega}\right)^{3/2}.}$$

Oscillator Wavefunctions and Heisenberg Motion

The ground state satisfies $a|0\rangle = 0$. In position space,

$$a = \sqrt{\frac{m\omega}{2\hbar}}X + \mathrm{i}\sqrt{\frac{1}{2m\omega\hbar}}P, \qquad 0 = \langle x|a|0\rangle = \left[\sqrt{\frac{m\omega}{2\hbar}}x + \sqrt{\frac{\hbar}{2m\omega}}\frac{\mathrm{d}}{\mathrm{d}x}\right]\langle x \mid 0\rangle.$$

Use the dimensionless coordinate $y = \sqrt{m\omega/\hbar}\,x$. With $\varphi_0(y)$ normalized using $\mathrm{d}y$,

$$\left(y + \frac{\mathrm{d}}{\mathrm{d}y}\right)\varphi_0 = 0, \quad \frac{\mathrm{d}\varphi_0}{\varphi_0} = -y\ \mathrm{d}y, \quad \varphi_0(y) = \pi^{-1/4}\mathrm{e}^{-y^2/2}.$$

Equivalently, the normalized position-space wavefunction is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\mathrm{e}^{-m\omega x^2/(2\hbar)}.$$

Raising generates all higher wavefunctions:

$$\varphi_n(y) = \frac{1}{\sqrt{n!}}\langle y|(a^\dagger)^n|0\rangle = \frac{1}{\pi^{1/4}\sqrt{2^n n!}}\left(y - \frac{\mathrm{d}}{\mathrm{d}y}\right)^n \mathrm{e}^{-y^2/2},$$
$$\psi_n(x) = \left(\frac{m\omega}{\hbar}\right)^{1/4}\varphi_n(y).$$

Clarification. The source mixes the x- and y-normalization constants and prints a plus sign in the raising derivative. The two normalizations are separated above; $a^\dagger = (y - \mathrm{d}/\,\mathrm{d}y)/\sqrt{2}$.

Classical oscillator motion

Hamilton's equations are $\dot{p} = -m\omega^2 x$ and $\dot{x} = p/m$. For a linear combination,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\alpha x + \beta p) = -m\omega^2\beta x + \frac{\alpha}{m}p.$$

Demanding proportionality imposes $\alpha/(-m\omega^2\beta) = \beta/(\alpha/m)$, hence $\alpha/\beta = \pm \mathrm{i}m\omega$. Choosing $C = m\omega x + \mathrm{i}p$ gives

$$\dot{C} = -\mathrm{i}\omega C, \qquad \dot{C}^* = \mathrm{i}\omega C^*, \qquad C(t) = C(0)\mathrm{e}^{-\mathrm{i}\omega t}.$$

Quantum oscillator motion

For $H = \hbar\omega(a^\dagger a + 1/2)$,

$$\frac{\mathrm{d}a}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}[a, H] = -\mathrm{i}\omega a, \qquad a(t) = a(0)\mathrm{e}^{-\mathrm{i}\omega t},$$
$$\frac{\mathrm{d}a^\dagger}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\left[a^\dagger, H\right] = \mathrm{i}\omega a^\dagger, \qquad a^\dagger(t) = a^\dagger(0)\mathrm{e}^{\mathrm{i}\omega t}.$$